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Embedding Geometric Heuristics for Improving Retrieval-Augmented Generation

Budget-Aware Search in the Embedding Space Guided by an LLM Judge

Research Proposal

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Abstract

In Retrieval-Augmented Generation (RAG), the quality of the answer is bounded by the quality of retrieval: if the relevant document is not retrieved, the generator cannot use it. Dense retrievers rank documents by embedding similarity to the query. They are cheap, but in specialised domains they miss the target document on a non-trivial fraction of queries. LLM judges score relevance much more accurately, but every call is expensive, and their scores are coarse: many documents receive the same top score.

This research treats retrieval with an LLM judge as a *budget-constrained sequential search problem*. Embedding distance is a cheap surrogate, the judge is an expensive oracle, and the policy chooses which document to judge next so as to reach the target in as few calls as possible. Our preliminary study used 196 PubMedQA questions over a shared pool of 5,035 documents drawn from a full-PubMed dense index, with all ≈1M question–document pairs scored by Gemini 2.5 Flash-Lite. It showed that (i) the judge's score is a reliable but coarse proxy for proximity to the target: its correlation with distance is negative in every query and ROC-AUC is 0.997, but a median of 18 documents per query share the top score; (ii) queries are geometric outliers with respect to the documents; and (iii) the centroid of the judge-approved documents is 44% closer to the target than the query is, yet static use of this centroid improves top-1 accuracy by at most one percentage point over the dense retriever (70.4%→71.4%), and naive online steering towards it increases cost.

Building on these findings, we will develop and evaluate (a) judge-guided search policies that invoke the expensive search only when the retriever is likely to have failed, use repulsive feedback from relevant-but-non-target documents, and localise the target by trilateration from judge scores; (b) finer-grained judging that breaks score ties; and (c) a characterisation and early prediction of retrieval failures. Policies will be compared by their cost–accuracy curves with dense retrieval, re-ranking, pseudo-relevance feedback and HyDE, on biomedical and general-domain benchmarks.

Keywords: retrieval-augmented generation, dense retrieval, embedding geometry, LLM-as-a-judge, budget-constrained search, biomedical question answering.

Contents

Abstract **ii**

1 Introduction and Background **1**

2 Problem Definition and Research Objectives **2**
2.1 Problem Definition 2
2.2 Research Objectives 2

3 Preliminary Results **3**
3.1 The Judge Is a Strong but Coarse Signal (RQ1) 3
3.2 Geometry of the Query–Document Space (RQ2) 3
3.3 Static and Adaptive Policies (H2) 4

4 Research Plan **5**
4.1 Methodology 5
4.2 Expected Contributions 6
4.3 Time Schedule 6

Bibliography **7**

1. Introduction and Background

Retrieval-Augmented Generation (RAG) grounds a large language model (LLM) in documents retrieved from an external corpus (Lewis et al., 2020; Gao et al., 2023b). Its answers are bounded by its retriever: if the document that answers the question is not retrieved, the generator cannot use it and may be misled by irrelevant context (Shi et al., 2023).

Two retrieval signals are available, with opposite strengths. *Dense retrievers* rank documents by the cosine similarity of their embeddings to the query embedding (Karpukhin et al., 2020; Reimers and Gurevych, 2019). This is almost free with a vector index (Malkov and Yashunin, 2020), but on our biomedical benchmark the top document is correct for only 70% of the questions. *LLM judges* rate relevance in close agreement with human assessors (Thomas et al., 2024). Yet every judgement is a costly LLM call, and labels on a small integer scale give many documents the same top score (Zhuang et al., 2024). Existing methods use the expensive signal passively. Re-rankers re-order a fixed top-k list (Nogueira and Cho, 2019; Sun et al., 2023): they pay k calls even when the retriever is right and cannot reach a target outside the list. Query rewriting and pseudo-relevance feedback move the retrieval anchor without true feedback (Rocchio, 1971; Gao et al., 2023a). Adaptive RAG detects *that* retrieval failed, not *where* to look next (Yan et al., 2024). Moreover, embedding spaces are anisotropic (Ethayarajh, 2019), and two-tower models can place their two input types in separate regions (Liang et al., 2022), so a question may lie far from the documents that answer it.

This research, developed in our supervision meetings, treats *judge scores as feedback that steers a search through the embedding space.* Each judged document indicates where the target lies, as in the word game Semantle or in trilateration from known anchors. Retrieval with a judge thus becomes the classical problem of optimising an expensive evaluation function with a cheap surrogate under a budget, as in best-first search (Pearl, 1984) and Bayesian optimisation (Shahriari et al., 2016). To our knowledge, no prior work studies the LLM judge as an expensive oracle inside a sequential search over the embedding space, with the number of judge calls as the cost and the query–document geometry as the explanation. This is the gap we address.

2. Problem Definition and Research Objectives

2.1 Problem Definition

Let D be a corpus, ϕ an embedding onto the unit sphere and $\delta(x, y) = 1 - \langle \phi(x), \phi(y) \rangle$ the cosine distance. A query q has a target document $GT(q) \in D$. A cheap *surrogate* $f'(d) = \delta(q, d)$ and an expensive *judge* $J(q, d) \in \{0.1, \ldots, 1.0\}$, which costs one LLM call, are available. A *search policy* π holds the judged history $H_t = \{(d_i, J(q, d_i))\}_{i \leq t}$ and chooses the next document to judge, $d_{t+1} = \pi(q, H_t)$: given the documents scored so far, *which document should the LLM check next?* The main metric is the **cost to find the target** $N_\pi(q)$, the number of judge calls until $GT(q)$ is judged. Secondary metrics are success within a budget and the accuracy of the returned document. A fixed ranking function f yields a static best-first policy whose cost is the target's rank. The standard retriever is $f = \delta(q, \cdot)$, which we call **B1**. Since J is pre-computed for all query–document pairs, any policy can be replayed offline at no LLM cost.

2.2 Research Objectives

The objective is to develop judge-guided search policies that find the target with fewer LLM calls than B1, especially where B1 fails, and to explain their behaviour through the geometry of the embedding space. **RQ1:** How reliable is the judge as a proximity signal, and how can ties at its top score be broken? **RQ2:** Which geometric properties explain retrieval failures? **RQ3:** Can policies that exploit judge feedback and geometry, invoked only when failure is likely, beat B1's cost? **RQ4:** Do the findings generalise across embedders, judges and datasets, and improve end-to-end RAG?

Hypotheses. **H1** Queries lie further from documents than documents from each other (supported). **H2** It is best to search mostly by distance to the query and a little by distance to the centroid of relevant documents (marginal gain only). **H3** Judge scores fall monotonically with distance to the target (supported, with ties). **H4** A *negative* weight on the centroid of relevant-but-non-target documents lowers the cost on failed queries. **H5** Starting the search only when the first document is not the target preserves B1's cost on easy queries. **H6** Trilateration from calibrated, finer-grained judge scores localises the target better than any single anchor.

3. Preliminary Results

Setup. On PubMedQA (Jin et al., 2019) with its own 272K-abstract corpus, the dense retriever (`all-mpnet-base-v2`, 768 dimensions; Song et al., 2020) ranked the target first for 93.5% of the questions, which leaves too little headroom. Following the supervisors' recommendation, we built a harder benchmark. We embedded the full PubMed corpus distributed with MedRAG (≈23.9M snippets; Xiong et al., 2024) on a cloud GPU instance. We then interleaved the 100 nearest neighbours of each of 199 questions and of each question's target into a shared pool of 5,035 documents. Every question–document pair (1,000,174 pairs) was scored by Gemini 2.5 Flash-Lite with a rubric on a 1–10 scale, normalised to 0.1–1.0. On this pool B1 ranks the target first for 70.4% of the 196 analysable questions, so it fails on 58.

3.1 The Judge Is a Strong but Coarse Signal (RQ1)

Per query, the judge's score correlates negatively with the distance from the document to the target: the median Spearman correlation is -0.29 and the median Pearson correlation is -0.50. The correlation is negative and significant for *every* query, and as a detector of the target the judge reaches ROC-AUC 0.997. The judge effectively uses six levels, and the mean distance to the target falls monotonically across them, from 0.89 at score 0.1 to 0.22 at score 1.0. The score is therefore usable as an ordinal "distance gauge". Its resolution at the top is poor, however: a median of 18 documents per query share the score 1.0 with the target. Moreover, on 46 of the 58 failed queries the judge also rates B1's wrong first document 1.0, so the current judge rarely flags a retrieval failure.

3.2 Geometry of the Query–Document Space (RQ2)

We drew balls of radius $\delta(q, GT)/2$ around each query and around its target. The query ball contained no document for all 196 queries, while the target ball was populated for 23% of them (38% of the failed ones). Queries are geometric outliers (H1). The centroid of the documents the judge rates 1.0 is 44% closer to the target than the query (mean distance 0.097 against 0.173), and it is closer for 92% of the queries, with a larger gain on failed ones. Its ball is smaller than the query's, yet it holds three times more documents. Control experiments showed that random points near the target

capture almost nothing, whereas any real document closer to the target than the query captures as many documents as the centroid. Documents are therefore concentrated around the target, and the query lies outside this region. Finally, the query–centroid–target triangle is nearly isosceles with its apex at the query ($\angle q \approx 46°$): the centroid points towards the *cluster* of relevant documents, not specifically towards the target.

3.3 Static and Adaptive Policies (H2)

Table 3.1 compares B1 with policies that mix in the centroid, first as an *oracle* (computed from all judge scores) and then *online* (from the documents judged so far). Sanity checks confirmed that B1 equals the RAG retriever, that the hybrid converges to B1 as $\alpha \to 1$, and that a centroid tie-breaker never fires, because cosine distances do not tie.

Table 3.1: Judge calls until the target is found (mean), on all 196 queries and on the 58 failed ones. Hybrid: $\alpha \,\mathrm{sim}(q, d) + (1 - \alpha)\,\mathrm{sim}(C, d)$.

Policy	Top-1 (%)	Calls	Calls (failed)
B1: distance to query (RAG)	70.4	**3.29**	**8.72**
Oracle centroid only	14.8	9.87	14.31
Oracle hybrid, $\alpha = 0.93$	**71.4**	3.35	8.93
Oracle hybrid, $\alpha = 0.70$	67.9	3.71	9.86
Top-10, re-ranked by oracle centroid	17.9	–	–
Online hybrid, $\alpha = 0.9$ / 0.5	70.4 / 70.4	3.35 / 4.10	8.93 / 11.47

Even with oracle information, the best hybrid gains one point at top-1 and changes the target's rank in only 16 queries (6 better, 10 worse). Online attraction towards judge-approved documents *increases* the cost, because it leads the search into the dense cluster of relevant-but-non-target documents. Judging B1's top-5 and returning the best (ties broken by distance) improves top-1 only from 73.8% to 75.4%, at five calls per query. Three conclusions follow, and they match the action items of our last meeting (12.10.2026): the search should run only when B1 is likely to have failed; it should push *away* from relevant-but-non-target documents rather than towards them; and it should exploit *graded* feedback, which requires a finer-grained judge.

4. Research Plan

4.1 Methodology

WP1 – Benchmark and evaluation protocol. We will extend the benchmark to the 1,000 expert-annotated PubMedQA questions with the same pool recipe, so that B1 still succeeds on 60–70% of them; documents outside the pre-scored pool will be scored on demand and cached. A power analysis based on the preliminary variances will set the number of failed queries needed (Sakai, 2016). Policies are reported on all queries and separately on the failed ones, with paired significance tests (Smucker et al., 2007). Baselines: B1 and BM25 (Robertson and Zaragoza, 2009); cross-encoder and LLM re-ranking of the top-k (Nogueira and Cho, 2019; Sun et al., 2023); pseudo-relevance feedback (Rocchio, 1971; Yu et al., 2021); and HyDE (Gao et al., 2023a).

WP2 – Characterising failure (RQ2). We will analyse in depth every query on which B1 is not perfect and compare it with the successful ones, using geometric features (distance to the target, off-axis displacement, local density) and judge features (score of the first document, size of the top-score group). From features available at query time we will train a failure predictor, compare it with query-performance predictors (Carmel and Yom-Tov, 2010), and use it to trigger the gated policies of WP4.

WP3 – Finer-grained judging (RQ1). To break ties we will compare a sharper rubric, a finer scale, expected relevance computed from label token probabilities (Zhuang et al., 2024), pairwise comparison applied only *within* the tied top group, and a stronger judge. Each variant is measured by tie size, discrimination and cost per call.

WP4 – Judge-guided search policies (RQ3). *(a) Gated search (H5):* judge B1's first document and stop if it is accepted; otherwise, switch to an adaptive policy. *(b) Repulsive feedback (H4):* select the next document by

$$s_t(d) = \alpha \,\mathrm{sim}(q, d) + \beta \,\mathrm{sim}(C_t^+, d) - \gamma \,\mathrm{sim}(C_t^-, d), \tag{4.1}$$

where C_t^+ is the centroid of the documents rated high but rejected as the target and C_t^- is the centroid of the documents rated low. The preliminary policy is $\beta > 0, \gamma = 0$; we will study $\beta < 0$, the negative weight from our last meeting, and Rocchio's $\gamma > 0$. *(c) Trilateration (H6):* the original idea of the project. We calibrate a monotone map g from judge score to distance, judge a few anchors chosen by farthest-point sampling near the query, and estimate the target as a weighted centroid or as $\hat{x} = \arg\min_x \sum_i (\delta(x, d_i) - g(J_i))^2$, restricted to the span of q and the judged documents. We then retrieve the nearest unjudged neighbour of $\hat{x}$ and repeat until a score threshold, stagnation or the budget is reached.

WP5 – Generalisation and end-to-end evaluation (RQ4). We will repeat the geometric analysis and the policy comparison with other embedders (Jin et al., 2023) and judges, and on BioASQ (Tsatsaronis et al., 2015) and MS MARCO (Nguyen et al., 2016). The best policies will be integrated into the RAG pipeline and evaluated on answer accuracy and LLM cost, following the MIRAGE protocol (Xiong et al., 2024).

4.2 Expected Contributions

The research is expected to contribute (1) a formulation of LLM-judged retrieval as budget-constrained sequential search, with an offline evaluation protocol over a large pre-scored pool; (2) an empirical account of the query–document geometry that explains why simple centroid heuristics fail; (3) judge-guided search policies and a failure predictor that reduce the number of LLM calls needed to recover from retrieval failures; and (4) practical guidelines on when, and how much, to spend on an LLM judge in a RAG system.

4.3 Time Schedule

Table 4.1: Work plan by semester (S1: Oct 2026–Mar 2027, S2: Apr–Sep 2027, S3: Oct 2027–Mar 2028, S4: Apr–Sep 2028).

Task	S1	S2	S3	S4
WP1: benchmark extension, power analysis	•			
WP2: failure characterisation and prediction	•	•		
WP3: finer-grained judging	•	•		
WP4: judge-guided search policies		•	•	
WP5: generalisation and end-to-end evaluation			•	•
Papers (judge and geometry; search policies)		•	•	
Thesis writing				•

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תקציר

מערכות יצירה מועשרת-אחזור (Retrieval-Augmented Generation, RAG) מעגנות את תשובות מודל השפה (LLM) במסמכים שנשלפים ממאגר ידע חיצוני, ולכן איכות התשובה חסומה מלמעלה באיכות האחזור: אם המסמך הרלוונטי לא אוחזר, המודל אינו יכול להשתמש בו. עומדים לרשותנו שני סוגים של אות אחזור. מאחזרים צפופים מדרגים מסמכים לפי הדמיון בין הייצוג הווקטורי (embedding) שלהם לבין ייצוג השאילתה. הם זולים, אך בתחומים מתמחים הם מחטיאים את מסמך המטרה בחלק לא זניח מהשאילתות. שופטי LLM מעריכים רלוונטיות בדיוק רב בהרבה, אך כל קריאה אליהם יקרה, והציונים שלהם גסים: מסמכים רבים מקבלים את אותו ציון מרבי.

במחקר זה אנו מנסחים אחזור בעזרת שופט LLM כבעיית *חיפוש סדרתי תחת תקציב*. המרחק במרחב הייצוגים משמש פונקציית קירוב זולה, והשופט משמש אורקל יקר. בכל צעד מדיניות החיפוש בוחרת איזה מסמך לשלוח לשופט, במטרה להגיע למסמך המטרה במספר קטן ככל האפשר של קריאות. במחקר המקדים בדקנו 196 שאלות מ-PubMedQA מול מאגר משותף של 5,035 מסמכים שנשלפו מאינדקס צפוף של PubMed המלא, וכל כמיליון זוגות השאלה–מסמך דורגו על ידי Gemini 2.5 Flash-Lite. המחקר המקדים ביסס שלוש עובדות. (א) ציון השופט הוא קירוב אמין אך גס לקרבה למסמך המטרה: המתאם עם המרחק שלילי בכל השאילתות ו-ROC-AUC שווה ל-0.997, אך בחציון 18 מסמכים לשאילתה חולקים את הציון המרבי. (ב) השאילתות הן נקודות חריגות גיאומטרית ביחס למסמכים. (ג) מרכז הכובד (צנטרואיד) של המסמכים שהשופט אישר קרוב למסמך המטרה ב-44% יותר מהשאילתה, אך שימוש סטטי בו משפר את דיוק המקום הראשון בנקודת אחוז אחת לכל היותר (מ-70.4% ל-71.4%), והטיה נאיבית ומקוונת של החיפוש לכיוונו מייקרת את החיפוש.

על בסיס ממצאים אלה נפתח ונבחן: (א) מדיניות חיפוש מונחית שופט, שמופעלת רק כאשר סביר שהמאחזר נכשל, משתמשת במשוב דוחה ממסמכים רלוונטיים שאינם המטרה, וממקמת את המטרה בשיטת טרילטרציה מציוני השופט; (ב) שיפוט ברזולוציה גבוהה יותר, שמפרק שוויונות בציון; (ג) אפיון וחיזוי מוקדם של כישלונות אחזור. המדיניות תושווה באמצעות עקומות עלות–דיוק לאחזור צפוף, לדירוג מחדש בעזרת cross-encoder ובעזרת LLM, למשוב רלוונטיות מדומה ול-HyDE, על מאגרים ביו-רפואיים ועל מאגרים מתחום כללי.

מילות מפתח: יצירה מועשרת-אחזור, אחזור צפוף, גיאומטריה של מרחב הייצוגים, LLM כשופט, חיפוש תחת תקציב, מענה על שאלות ביו-רפואיות.

תוכן עניינים

תקציר (באנגלית) ii

1. מבוא ורקע 1

2. הגדרת הבעיה ומטרות המחקר 2

3. תוצאות מקדימות 3

4. תוכנית המחקר 5

ביבליוגרפיה 7

אוניברסיטת בן-גוריון בנגב
הפקולטה למדעי המחשב והמידע
המחלקה למערכות מידע

יוריסטיקות גיאומטריות במרחב הייצוגים הווקטוריים לשיפור יצירה מועשרת-אחזור (RAG)

חיפוש מונחה שופט LLM תחת תקציב קריאות

הצעת מחקר

בר מרדכי דמרי
ת"ז: 208915751

בהנחיית:
פרופ' רוני שטרן פרופ' רמי פוזיס

13 באוקטובר 2026